ENGINEERING ECONOMICS (CE 655)

Chapter 1: Introduction

Questions & Answers (TU Past Exam Papers)

Q1. Define cash flow diagram. Why study of economics is essential for engineers? [2082 Kartik, 4 Marks]

Answer:

(A) Cash Flow Diagram

A cash flow diagram is a graphical representation of all cash transactions (both receipts and disbursements) plotted on a time scale. It is a visual tool used in engineering economics to depict the timing and magnitude of all cash flows associated with a project or investment.

Components of a Cash Flow Diagram:

- Horizontal Axis (Time Line): Represents time, usually in years or other uniform time periods. The present time is marked as '0', and future periods are marked as 1, 2, 3, ... n.
- Upward Arrows (Cash Inflows/Receipts): Represent positive cash flows such as revenues, salvage values, or deposits received. Arrow height is proportional to magnitude.
- Downward Arrows (Cash Outflows/Disbursements): Represent negative cash flows such as initial investments, operating costs, or loan repayments.
- Arrow Magnitude: The length/height of each arrow represents the monetary amount of the cash flow.

Example: If a project has an initial investment of Rs. 1,00,000 (at t=0), annual revenues of Rs. 30,000 (at end of each year), annual operating cost of Rs. 5,000, and a salvage value of Rs. 20,000 at the end of year 5, the cash flow diagram would show:

- A downward arrow of Rs. 1,00,000 at t = 0
- Net upward arrows of Rs. 25,000 at t = 1, 2, 3, 4
- Net upward arrow of Rs. 45,000 at t = 5 (25,000 + 20,000)

(B) Why Study of Economics is Essential for Engineers

Engineers are decision-makers who design, build, and operate technical systems that require significant financial resources. The study of economics is essential for engineers for the following reasons:

1. Investment Decision Making: Engineers often recommend or make decisions on capital investments. Economic analysis enables them to evaluate whether investments will provide adequate returns.
2. Comparison of Alternatives: Multiple technical solutions often exist for a problem. Engineers must compare these alternatives not only technically but also economically.
3. Cost Optimization: Engineers can apply economic principles to minimize total life-cycle costs of projects, considering both capital and operating costs.
4. Justifying Engineering Decisions: Economic analysis provides quantitative justification for engineering proposals to management and stakeholders.
5. Resource Allocation: With limited budgets, engineers must allocate resources efficiently. Economic analysis guides optimal allocation.

6. Time Value of Money: Engineers deal with multi-year projects. Understanding time value of money helps them make correct comparisons across different time periods.
7. Project Feasibility: Before undertaking any project, engineers must determine its economic feasibility to ensure it is worth pursuing.
8. Profitability Analysis: Engineers must assess whether projects will generate sufficient profit to meet the organization's financial objectives.

In summary, engineers who lack economic knowledge may design technically perfect systems that are economically unviable, wasting resources and potentially causing financial losses to their organizations.

Q2. Explain the principles of Engineering Economics. [2081 Chaitra, 4 Marks; 2080 Chaitra, 4 Marks; Multiple Years]

Answer:

Engineering economics is guided by several fundamental principles that govern economic decision-making in engineering contexts. These principles were formulated by Sullivan, Wicks, and Koelling. The key principles are:

Principle 1: Develop the Alternatives

The choice is among alternatives. Always develop and identify all feasible alternatives before making a decision. A decision is only as good as the best alternative being considered. Never limit analysis to a single option; always compare against 'do nothing' and other viable solutions.

Principle 2: Focus on the Differences

Only the differences in expected future outcomes among alternatives are relevant to their comparison and should be considered in the decision. Common costs and benefits that are the same for all alternatives should be ignored as they do not affect the relative comparison.

Principle 3: Use a Consistent Viewpoint

Prospective outcomes of alternatives should be consistently developed, identified, and observed from a defined viewpoint — typically the owner's perspective. Mixing viewpoints (owner vs. society vs. government) in the same analysis leads to inconsistent and unreliable results.

Principle 4: Use a Common Unit of Measure

Converting all outcomes to a common monetary unit (such as rupees) makes comparisons straightforward and objective. Non-monetary outcomes should be quantified where possible, or separately acknowledged if they cannot be monetized.

Principle 5: Consider All Relevant Criteria

The selection of the preferred alternative requires use of a criterion (or several criteria). Monetary profitability is important, but non-monetary factors such as safety, environmental impact, social consequences, and regulatory requirements must also be considered in the final decision.

Principle 6: Make Uncertainty Explicit

Uncertainty is inherent in projecting future outcomes. Estimates about revenues, costs, useful lives, and other parameters involve uncertainty. This uncertainty should be explicitly recognized, analyzed through sensitivity analysis, scenario analysis, or probability methods, rather than ignored.

Principle 7: Revisit Your Decisions (Adaptive Management)

Improve decision making by using a retrospective (post-project) analysis of results as compared to the estimates. Learning from past decisions helps refine future economic analyses and improves the quality of future decisions.

Principle 8: Use the Systems Viewpoint

Consider the broader impact of each alternative on the entire system, not just the component or subsystem under study. An engineering decision that optimizes one part may create problems elsewhere in the system.

Principle 9: Adopt the Time Value of Money

Money has time value — a rupee today is worth more than a rupee in the future due to its earning potential. All economic analyses must properly account for the time value of money using interest rates and present/future worth calculations.

Q3. Define Engineering Economics. Briefly explain the scope of Engineering Economics with an appropriate example. [2080 Chaitra, 1+3 Marks]

Answer:

Definition of Engineering Economics

Engineering economics (also called Engineering Economy) is the application of economic principles and quantitative methods to the analysis, comparison, and selection of engineering projects, systems, and processes. It provides a framework for making systematic and rational economic decisions in engineering.

It is defined as: 'The systematic evaluation of the economic merits of proposed solutions to engineering problems.'

Scope of Engineering Economics

Engineering economics covers a broad range of topics relevant to engineering decision-making:

1. Time Value of Money: Understanding that money changes in value over time due to interest and inflation. This includes calculating present worth, future worth, and annual worth of cash flows.
2. Interest and Equivalence: Calculating and comparing cash flows at different points in time using interest formulas (P/F, F/P, A/F, A/P, gradient factors, etc.).
3. Project Evaluation Methods: Using methodologies such as Net Present Value (NPV/PW), Internal Rate of Return (IRR), Annual Worth (AW), and Benefit-Cost Ratio (BCR) to evaluate project feasibility.
4. Comparison of Alternatives: Comparing mutually exclusive, independent, and contingent projects to determine the most economically attractive option.
5. Depreciation: Analyzing the reduction in value of engineering assets over time using methods like Straight Line, Declining Balance, SOYD, and MACRS.
6. Replacement Analysis: Determining the optimal time to replace aging equipment (defender vs. challenger analysis).
7. Risk and Sensitivity Analysis: Evaluating the impact of uncertainty on project outcomes through sensitivity analysis, scenario analysis, and decision trees.
8. Inflation Analysis: Accounting for the effects of inflation on engineering project cash flows using actual dollar and constant dollar analyses.
9. Income Taxes: Understanding how corporate income taxes affect after-tax cash flows and project profitability.

Example:

A civil engineering firm is considering building a new bridge. The initial cost is Rs. 50 Crores. It will generate annual toll revenue of Rs. 8 Crores for 20 years. The annual maintenance cost is Rs. 1 Crore. Engineering economics is used to:

- Calculate the Net Present Value (NPV) of the project at a given interest rate to determine if it is economically viable.
- Compare this bridge design against alternative bridge designs with different costs and revenue potential.
- Determine the Payback Period and Internal Rate of Return.
- Analyze sensitivity to traffic volume (which affects toll revenue) uncertainty.

Q4. Why do engineers need knowledge of economics in decision making process? Explain briefly about the principles of engineering economics. [2081 Ashwin, 1+3 Marks]

Answer:

Why Engineers Need Knowledge of Economics

Engineers are not merely technical professionals — they are decision makers who manage resources. Knowledge of economics is essential for engineers in the decision-making process for these reasons:

1. Technical Excellence Alone is Insufficient: A technically perfect solution that is economically unviable will not be implemented. Engineers must balance technical performance with cost-effectiveness.
2. Capital Investment Decisions: Engineers recommend and justify major capital investments. Without economic knowledge, they cannot properly evaluate whether an investment will yield adequate returns.
3. Life Cycle Cost Analysis: Engineering decisions have long-term financial implications. Engineers need to consider all costs over the life of a project — initial, operating, maintenance, and disposal costs.
4. Communication with Management: Engineers must communicate with financial managers and executives using economic language and metrics (NPV, IRR, payback period).
5. Resource Optimization: In resource-constrained environments, engineers must prioritize investments using economic criteria to maximize value.
6. Regulatory and Public Interest: In civil and public works projects, engineers must justify expenditures on behalf of society using benefit-cost analysis.

For the principles of engineering economics, please refer to Q2 above (Develop Alternatives, Focus on Differences, Consistent Viewpoint, Common Unit, All Criteria, Explicit Uncertainty, Revisit Decisions, Systems Viewpoint, Time Value of Money).

Q5. "The study of economics is important for engineers". Justify the statement with a suitable example. [2079 Chaitra, 4 Marks]

Answer:

The statement that 'the study of economics is important for engineers' is absolutely justified because engineering and economics are inseparably linked in practice.

Justification:

1. Every Engineering Decision has Economic Consequences: Whether selecting a material, choosing a structural design, or deciding on a construction method, engineers make decisions that carry significant monetary implications.
2. Limited Resources: Society has limited financial, material, and human resources. Engineers must use these resources wisely, which requires economic analysis to optimize their utilization.

3. Project Viability: Not every technically feasible project is economically viable. Engineers must be able to screen projects on economic grounds before committing resources.
4. Cost Management: Engineers directly influence construction and operating costs through design decisions. Understanding economics helps minimize unnecessary costs.
5. Return on Investment: Organizations invest in engineering projects expecting returns. Engineers must ensure that designed systems generate the expected financial returns.
6. Long-term Planning: Engineering assets last for decades. Economic analysis over the life cycle of an asset requires understanding inflation, depreciation, and time value of money.

Example:

Consider a water supply project for a municipality. Two options are available:

- Option A: Rs. 1 Crore initial cost, Rs. 5 Lakhs/year O&M, life 20 years
- Option B: Rs. 70 Lakhs initial cost, Rs. 9 Lakhs/year O&M, life 20 years

Without economic analysis, an engineer might choose Option B because of lower initial cost. However, by computing the Annual Equivalent Cost (AEC) at 10% interest:

- AEC of A = 1,00,00,000 (A/P, 10%, 20) + 5,00,000 = Rs. 16.75 Lakhs/year
- AEC of B = 70,00,000 (A/P, 10%, 20) + 9,00,000 = Rs. 17.22 Lakhs/year

The economic analysis reveals that Option A is actually more economical over the life of the project, despite having a higher initial cost. This demonstrates how engineering economics enables engineers to make better, more informed decisions.

Q6. Explain the role of engineers in decision making in investment. [2079 Ashwin, 4 Marks; 2077 Chaitra, 4 Marks; 2070 Magh, 4 Marks]

Answer:

Engineers play a vital and multifaceted role in investment decision-making processes within organizations. Their technical expertise combined with economic analysis skills makes them key contributors at every stage of the investment decision cycle.

Roles of Engineers in Investment Decision Making:

1. Problem Identification and Problem Definition: Engineers identify technical problems and opportunities that may require capital investment. They define the scope and boundaries of the problem clearly, which is the foundation for economic analysis.
2. Developing Alternatives: Engineers generate creative, technically feasible alternatives for solving identified problems. The quality of economic analysis depends directly on the quality of alternatives generated.
3. Data Collection and Estimation: Engineers collect technical data and prepare cost and benefit estimates for each alternative. This includes estimating initial costs, operating costs, maintenance costs, useful life, and salvage values.
4. Economic Analysis: Engineers apply economic analysis techniques (NPV, IRR, AW, B/C ratio, payback period) to evaluate and compare alternatives on an economic basis.
5. Risk Assessment: Engineers identify technical risks associated with each alternative and quantify their potential impact on project economics through sensitivity and scenario analyses.
6. Recommendation and Presentation: Engineers prepare and present investment proposals with supporting economic analysis to management for decision-making. They must communicate complex technical and economic information clearly.
7. Implementation and Monitoring: After approval, engineers implement the selected alternative and monitor actual performance against economic projections, providing feedback for future decisions.

8. Post-project Evaluation: Engineers conduct post-completion audits comparing actual vs. estimated costs and benefits, improving the quality of future investment analyses.

The Engineering Decision Making Process:

- Step 1: Identify and define the problem
- Step 2: Establish the goal or objective
- Step 3: Collect relevant data and information
- Step 4: Identify all feasible alternatives
- Step 5: Analyze each alternative economically
- Step 6: Compare alternatives using common criteria
- Step 7: Select the best alternative
- Step 8: Implement the decision
- Step 9: Monitor results and learn

Engineers who combine strong technical knowledge with economic analysis skills are most effective in guiding sound investment decisions that maximize value for their organizations and society.

Q7. "Use the consistent view point" and "Make uncertainty explicit". Explain these two principles of engineering economics. [2078 Chaitra, 2+2 Marks]

Answer:

Principle: Use a Consistent Viewpoint

This principle states that when analyzing engineering alternatives, all outcomes and consequences must be identified, quantified, and compared consistently from a single, defined perspective or viewpoint throughout the analysis.

Why it is Important:

- Different stakeholders (owner, contractor, government, society) have different costs and benefits from the same project.
- Mixing viewpoints in the same analysis produces meaningless or misleading results.
- A consistent viewpoint ensures apples-to-apples comparison of alternatives.

Common Viewpoints in Engineering Economics:

- Owner's viewpoint: Focuses on costs and revenues to the project owner.
- Society's viewpoint: Considers all social costs and benefits (used in public sector analysis).
- Government viewpoint: Considers tax revenues, subsidies, and public expenditures.

Example:

When comparing two bridge designs for a government agency, the analysis must consistently use the same viewpoint (say, the government's perspective) for both alternatives. If one alternative's benefits are computed from the government's perspective but another's from the public's perspective, the comparison is invalid.

Principle: Make Uncertainty Explicit

This principle recognizes that in engineering economics, all future estimates (costs, revenues, useful life, salvage values) involve uncertainty. These uncertainties should be explicitly acknowledged and analyzed rather than hidden or ignored.

Why it is Important:

- Economic analyses rely on estimates of future events, which are inherently uncertain.

- Ignoring uncertainty can lead to overly optimistic or pessimistic project evaluations.
- Explicit uncertainty analysis helps decision-makers understand the range of possible outcomes.
- It improves the quality and reliability of engineering economic decisions.

Methods to Make Uncertainty Explicit:

- Sensitivity Analysis: Vary one parameter at a time (e.g., ±20% on investment) and observe the impact on decision metrics like NPV or IRR.
- Scenario Analysis: Analyze best case, worst case, and most likely case scenarios.
- Probability Analysis: Assign probability distributions to uncertain parameters and compute expected values.
- Decision Tree Analysis: Map out sequential decisions and their probabilistic outcomes.

Example:

If a project's annual revenue is estimated at Rs. 50,000 but could range from Rs. 40,000 to Rs. 60,000, sensitivity analysis should be performed to show how the project's NPV changes across this range, rather than simply computing NPV at Rs. 50,000 alone.

Q8. Explain the importance of Engineering Economics for Engineers. [2078 Poush, 4 Marks]

Answer:

Engineering economics is critically important for engineers in both professional practice and academic preparation. The following points highlight its significance:

1. Economic Decision Making: Engineering decisions involving capital investments, equipment purchases, project selections, and resource allocation require systematic economic evaluation. Engineering economics provides the tools to make these decisions rationally.
2. Bridging Technical and Financial Worlds: Engineers work at the interface of technical systems and financial management. Engineering economics serves as the bridge between technical analysis and financial decision-making.
3. Maximizing Value: In competitive environments, organizations seek maximum value from their investments. Engineers trained in economic analysis can design and recommend solutions that maximize return on investment.
4. Cost Control: Understanding life-cycle costs and depreciation helps engineers design cost-effective systems and identify opportunities for cost reduction throughout a project's life.
5. Public Sector Analysis: Civil engineers frequently work on public infrastructure projects. Benefit-cost analysis, a key tool in engineering economics, is the standard method for evaluating public investments.
6. Regulatory Compliance: Many infrastructure projects require economic justification as part of regulatory approvals (environmental impact assessments, feasibility studies). Engineers must provide these economic analyses.
7. Entrepreneurship and Innovation: Engineers who understand economics are better equipped to evaluate new business ventures, start-ups, and innovative projects from a financial viability perspective.
8. Professional Competence: Engineering professional bodies around the world recognize economic analysis as a core engineering competency. It is a required subject in most engineering curricula.
9. Inflation and Tax Effects: Long-duration engineering projects are significantly affected by inflation and taxes. Engineering economics provides methods to correctly account for these effects.

10. Replacement and Maintenance Decisions: Engineers must determine when to replace aging equipment and whether to repair or replace assets. These decisions require economic analysis of defender vs. challenger options.

Q9. Describe different principles of engineering economics. What do you mean by sunk cost and opportunity cost? [2079 Jestha, 4+2 Marks]

Answer:

For the principles of engineering economics, refer to Q2 above. The key principles are: (1) Develop the Alternatives, (2) Focus on the Differences, (3) Use a Consistent Viewpoint, (4) Use a Common Unit of Measure, (5) Consider All Relevant Criteria, (6) Make Uncertainty Explicit, (7) Revisit Your Decisions, (8) Use the Systems Viewpoint, and (9) Adopt the Time Value of Money.

Sunk Cost

Definition: A sunk cost is a past cost that has already been incurred and cannot be recovered, regardless of any future decision or action taken. Since sunk costs cannot be changed by any future decision, they should not be included in economic analyses of future alternatives.

Characteristics of Sunk Costs:

- They are historical, past costs — already spent.
- They are irretrievable — cannot be recovered.
- They should NOT influence future decisions (to avoid 'sunk cost fallacy').

Example: A company has already spent Rs. 20 Lakhs on a project design. If the project is later found to be uneconomical, the Rs. 20 Lakhs is a sunk cost. The decision to continue or abandon the project should be based on future costs and benefits only, not on the Rs. 20 Lakhs already spent.

Opportunity Cost

Definition: Opportunity cost is the cost of the forgone next best alternative when a choice is made. It represents the value of the best alternative that is sacrificed when selecting one option over another.

Characteristics of Opportunity Cost:

- It represents the foregone benefit of the second-best alternative.
- It is implicit (not always directly visible in financial statements) but must be considered.
- It is relevant for comparing alternatives and for resource allocation decisions.

Example: A civil engineer has Rs. 10,00,000. Option A is to invest in Machine A expected to earn Rs. 2,00,000 per year. Option B is to invest in Machine B expected to earn Rs. 1,50,000 per year. If the engineer chooses Machine A, the opportunity cost is Rs. 1,50,000/year (what Machine B would have earned). The net advantage of Machine A is Rs. 50,000/year above the opportunity cost.

In replacement analysis, the opportunity cost of keeping the defender (existing equipment) is its current market value — which would have been received if the defender was sold.

Q10. "Knowledge of engineering economics helps in decision making process". Justify it by the principles of engineering economics. [2073 Magh, 4 Marks]

Answer:

Engineering economics provides systematic principles and tools that directly enable better decision making. Each principle of engineering economics contributes to improved decision quality:

1. Principle 'Develop the Alternatives' helps decision makers: By requiring that multiple options be developed and analyzed, this principle ensures that the best available option is identified and not overlooked. It prevents premature decisions based on limited options.

2. Principle 'Focus on the Differences' simplifies decision making: By directing attention only to differences between alternatives, it eliminates irrelevant information and focuses analysis where it matters — on what changes with each choice.

3. Principle 'Use Consistent Viewpoint' ensures valid comparisons: Analyzing all alternatives from the same perspective (owner, society, etc.) produces reliable, fair comparisons that lead to correct decisions.

4. Principle 'Use a Common Unit' enables objective comparison: Converting all outcomes to monetary terms allows decision makers to objectively rank alternatives and apply quantitative decision criteria.

5. Principle 'Consider All Criteria' leads to holistic decisions: By considering non-monetary factors alongside monetary ones, engineering economics ensures that decisions are not purely profit-driven but also account for safety, sustainability, and social impact.

6. Principle 'Make Uncertainty Explicit' leads to robust decisions: Analyzing how uncertainty affects outcomes allows decision makers to choose alternatives that are robust under varying conditions, reducing the risk of poor outcomes.

7. Principle 'Time Value of Money' ensures accurate financial comparison: All engineering projects extend over time. Accounting for time value of money ensures that cash flows in different periods are correctly compared.

In summary, without these principles, decisions would be based on incomplete information, inconsistent comparisons, and financial errors — leading to suboptimal resource use and economic losses.

Q11. Scarcity is an emerging issue in engineering field. How does the study of economics help engineers in decision making process? Discuss. [2070 Bhadra, 5 Marks]

Answer:

Scarcity in Engineering

Scarcity refers to the fundamental economic problem that resources (capital, materials, labor, energy, land) are limited while human wants and needs are unlimited. In engineering, scarcity manifests as:

- Limited budget for capital investments
- Shortage of skilled labor and materials
- Limited availability of land and natural resources
- Constrained energy resources
- Time constraints on project completion

How Economics Helps Engineers Address Scarcity:

1. Rational Resource Allocation: Engineering economics provides methods (such as incremental rate of return analysis) to allocate limited budgets among competing projects to maximize total benefit.

2. Prioritization of Projects: When not all desired projects can be funded, economic analysis (using NPV, IRR, or B/C ratios) helps rank projects to identify the most valuable ones for implementation.

3. Trade-off Analysis: Economics enables engineers to evaluate trade-offs between higher initial cost (using more durable but expensive materials) and lower long-term cost (less maintenance), making optimal choices given resource constraints.

4. Minimizing Waste: Cost minimization through value engineering and life-cycle cost analysis ensures that scarce resources are not wasted.

5. Demand Management: Understanding economic demand curves helps engineers design systems sized appropriately for expected demand rather than over-designing (wasting resources) or under-designing (insufficient capacity).
6. Substitute Analysis: When scarce resources are unavailable, economics helps evaluate substitute materials or methods and determine if the substitution is cost-effective.
7. Opportunity Cost Assessment: By explicitly considering opportunity costs, engineers ensure that scarce capital is directed to its most valuable use.

In essence, economics provides the analytical framework that helps engineers make the best possible decisions under conditions of resource scarcity, ensuring maximum value is extracted from available resources.

Q12. List out the principles of engineering economy. Explain the cash flow diagram. [2071 Magh, 1+3 Marks]

Answer:

Principles of Engineering Economy

The nine key principles of engineering economy are:

1. Develop the Alternatives
2. Focus on the Differences
3. Use a Consistent Viewpoint
4. Use a Common Unit of Measure
5. Consider All Relevant Criteria
6. Make Uncertainty Explicit
7. Revisit Your Decisions
8. Use the Systems Viewpoint
9. Adopt the Time Value of Money

Cash Flow Diagram

A cash flow diagram is a visual representation of the timing and magnitude of cash transactions over the life of a project or investment. It serves as a fundamental tool in engineering economics to organize and display financial data before performing calculations.

Rules for Drawing a Cash Flow Diagram:

1. Draw a horizontal line representing the time scale. Divide it into equal time periods (usually years). Mark 0 at the beginning (present time) and subsequent periods as 1, 2, 3, ... n.
2. Draw arrows perpendicular to the time line at the appropriate time periods. Upward arrows represent cash inflows (revenues, savings, salvage values). Downward arrows represent cash outflows (investments, costs, loan payments).
3. The length/height of each arrow should be proportional to the magnitude of the cash flow amount.
4. Label each arrow with its monetary value.
5. Specify the viewpoint (owner, borrower, lender) from which the diagram is drawn.

Conventions:

- End-of-period convention: Cash flows are assumed to occur at the end of each period unless stated otherwise.
- The perspective matters: A positive cash flow for the borrower is a negative cash flow for the lender.

Example:

Consider a loan of Rs. 50,000 taken now (t=0) to be repaid in 5 equal annual installments of Rs. 13,189 each at end of each year at 10% interest. From the borrower's perspective:

- t=0: Upward arrow of Rs. 50,000 (loan received — cash inflow)
- t=1 to t=5: Downward arrows of Rs. 13,189 each (repayments — cash outflow)

From the lender's perspective, the arrows are exactly reversed.

Q13. Define Engineering Economics. Why engineering economics is considered as important aspect for making decisions for engineers? Explain. [2076 Bhadra, 1+3 Marks]

Answer:

Definition

Engineering Economics is the branch of economics that is concerned with the application of economic principles and quantitative methods for making decisions related to engineering projects, systems, and processes. It provides the tools to analyze the economic merits of engineering alternatives and to determine the best course of action from an economic standpoint.

Alternative definitions: 'Engineering economics involves the systematic evaluation of the economic merits of proposed solutions to engineering problems.' (Sullivan et al.)

Importance of Engineering Economics for Engineers

Engineering economics is an essential discipline for engineers for the following key reasons:

1. Technical Feasibility vs. Economic Viability: A technically sound design must also be economically viable. Engineering economics helps verify that engineering solutions are not only technically workable but also financially justifiable.
2. Capital Budgeting: Engineers recommend capital expenditures for machinery, infrastructure, and technology. Economic analysis justifies these expenditures by demonstrating expected returns.
3. Comparative Analysis of Alternatives: Multiple solutions typically exist for any engineering problem. Engineering economics enables systematic comparison to identify the most cost-effective solution.
4. Life-Cycle Cost Analysis: Total project costs include not just initial costs but also operating, maintenance, and disposal costs over the entire life. Engineers must consider all these costs for complete economic evaluation.
5. Time Value of Money: Engineering projects span many years. Economic analysis accounts for the fact that money received or spent at different times has different values, ensuring accurate financial comparisons.
6. Stakeholder Accountability: Engineers must be accountable for the economic outcomes of their decisions. Economic analysis provides quantitative evidence to support and justify engineering recommendations.
7. Public Projects: For government and public infrastructure projects, engineers must conduct benefit-cost analysis to demonstrate that public funds are being utilized optimally.

Q14. Define opportunity cost. Why engineering economics is considered as important aspect for making decisions for engineers? [2075 Bhadra, 1+3 Marks]

Answer:

Opportunity Cost

Opportunity cost is defined as the economic value of the best alternative that is foregone or sacrificed when a decision is made to pursue a particular course of action. It represents what you give up when you choose one option over another.

In engineering economics: 'The opportunity cost of keeping the defender (old equipment) in service is its current market value — the amount that could have been received by selling it today.'

Examples:

- If a company invests Rs. 10 Lakhs in Project A earning 12% return, and the next best alternative is Project B earning 10%, the opportunity cost is the 10% return from Project B.
- If an engineer uses existing land (valued at Rs. 50 Lakhs) for a factory, the opportunity cost is the rental income that could have been earned by leasing the land.
- When a company retains old equipment worth Rs. 20 Lakhs (market value) rather than selling it, the opportunity cost is Rs. 20 Lakhs.

For the importance of engineering economics for engineers, please refer to Q13 above.

--- End of Chapter 1 Questions & Answers ---